

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued. . .]

- Appendix on s -wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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Contents

1	Phase transitions and SSB in EHM	3
1.1	The EHM symmetries	3
1.1.1	Lattice symmetries	3
1.1.2	Spin and charge symmetries	4
1.1.3	Particle-hole symmetry at half-filling	8
1.2	Symmetry breaking channels in real space	10
1.2.1	Local repulsion	11
1.2.2	Non-local attraction	12
1.3	The EHM in reciprocal space	14
1.4	Square lattice spatial symmetry channels	17
	Bibliography	21

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (??) and then apply the iHF procedure sketched in Sec. ?? to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;
2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;
3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other “phases” are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (??). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

1.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model

$$\hat{H}_{\text{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\text{EHM}}$. We start by the lattice symmetries.

1.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. A sketch of the lattice $\mathcal{L}$ is given in Fig. [Missing]. We indicate with L_x the number of sites in direction x , and L_y in direction y . The hamiltonian of Eq. (??) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as $T_1^{(x)}$, $T_1^{(y)}$. It holds:

$$T_n^{(\ell)} \supset T_{kn}^{(\ell)} \quad k \in \mathbb{N} \quad (1.1)$$

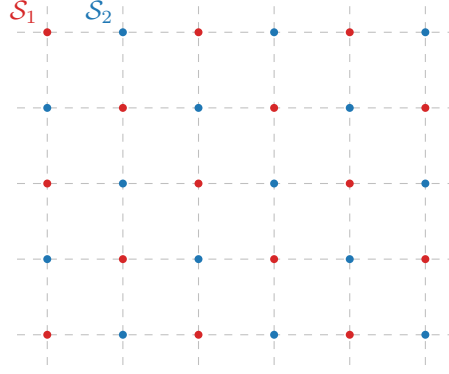


Figure 1.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n -sites shifts, thus, implies invariance also for kn -sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4 , containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$C_n \supset C_{n/k} \quad k, n/k \in \mathbb{N} \quad (1.2)$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$I_2 \equiv C_2 \quad (1.3)$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v, w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1, \mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \implies (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 1.1. Two sites are “connected” if linked by a non-zero hopping operator. Which is, if they are NNs.

1.1.2 Spin and charge symmetries

The conventional HM shows a global $U(2)$ symmetry, extended to $SU(2) \times SU(2)/\mathbb{Z}_2 = SO(4)$ symmetry at half-filling on a bipartite lattice [1], such as the square lattice. For the EHM, we limit to prove that we can identify a $U(2) \subset SO(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \rightarrow e^{-i\eta} \hat{c}_{i\sigma} \quad \hat{c}_{i\sigma}^\dagger \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger \quad \eta \in \mathbb{R} \quad (1.4)$$

To prove this, it suffices to note that

$$\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger e^{-i\eta} \hat{c}_{j\sigma'} = \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \quad (1.5)$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U, \hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (1.4), which is a pure phase shift by η . The relative group is $U^c(1)$, with a superscript “c” indicating “charge”.

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ ; local repulsion between opposite-spin densities; non-local attraction, independent of σ . If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\text{EHM}} - \mu \hat{N}$ must exhibit a global $SU(2)$ symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \quad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \quad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}_i^\alpha \equiv \frac{1}{2} \hat{\Psi}_i^\dagger \tau^\alpha \hat{\Psi}_i \quad [\hat{S}_j^\alpha, \hat{S}_j^\beta] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}_j^\gamma$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha = x, \beta = y$:

$$\begin{aligned} [\hat{S}_j^x, \hat{S}_j^y] &= \frac{1}{4} \left[\hat{\Psi}_j^\dagger \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}_j^\dagger \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j \right] \\ &= \frac{1}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, -i \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + i \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] - \frac{i}{4} \left[\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right] \\ &= \frac{i}{2} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(-\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} + \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\downarrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= i \hat{S}_j^z \end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}_{j\sigma'}^\dagger\} = \delta_{ij} \delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \quad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i \in \mathcal{L}} \hat{S}_i^\alpha \quad [\hat{S}^\alpha, \hat{S}^\beta] = i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \quad \hat{S}^+ \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \quad \hat{S}^- \equiv (\hat{S}^+)^\dagger$$

The EHM hamiltonian and the number operator commute with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [1, 2],

$$[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}] = 0 \quad [\hat{N}, \hat{\mathbf{S}}] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned} \hat{H}_V &= V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \\ &= V \sum_{\langle ij \rangle} \hat{\Psi}_i^\dagger \hat{\Psi}_i \hat{\Psi}_j^\dagger \hat{\Psi}_j \end{aligned} \quad (1.6)$$

having we used $\hat{\Psi}_i^\dagger \hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned} \hat{N} &= \sum_{i \in \mathcal{L}} \sum_{\sigma} \hat{n}_{i\sigma} \\ &= \sum_{i \in \mathcal{L}} \hat{\Psi}_i^\dagger \hat{\Psi}_i \end{aligned} \quad (1.7)$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp(-i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}) \quad \text{where } \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

Spinors transform as:

$$\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i$$

with $\mathcal{R}(\boldsymbol{\theta}) \in \text{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau}\right) \quad \text{where } \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \quad (1.8)$$

Let us prove this. Let $\hat{A} = i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

The first order of the expansion gives:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] &= i \sum_{\alpha} \theta_{\alpha} [\hat{S}^{\alpha}, \hat{\Psi}_i] \\
&= -\frac{i}{2} \sum_{\alpha} \theta_{\alpha} \tau^{\alpha} \hat{\Psi}_i \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i
\end{aligned}$$

We omitted some purely algebraic passages. The second order:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i]] &= [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, -\frac{i}{2}(\boldsymbol{\theta} \cdot \boldsymbol{\tau}) \hat{\Psi}_i] \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] \\
&= \frac{1}{4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i
\end{aligned}$$

and so on. Collecting all terms,

$$\begin{aligned}
\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^{\dagger} &= \hat{\Psi}_i - \frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i + \frac{1}{2! \times 4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i + \dots \\
&= \sum_{k=1}^{\infty} \frac{1}{k!} \left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \right)^k \hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i
\end{aligned}$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (1.8).

4. Then, the group acts on the product $\hat{\Psi}_i^{\dagger} \hat{\Psi}_i$ as

$$\begin{aligned}
\hat{R} \hat{\Psi}_i^{\dagger} \hat{\Psi}_i \hat{R}^{\dagger} &= \hat{R} \hat{\Psi}_i^{\dagger} \hat{R}^{\dagger} \hat{R} \hat{\Psi}_i \hat{R}^{\dagger} \\
&= \hat{\Psi}_i^{\dagger} \mathcal{R}^{\dagger} \mathcal{R} \hat{\Psi}_i \\
&= \hat{\Psi}_i^{\dagger} \hat{\Psi}_i
\end{aligned}$$

being $\mathcal{R} \in \text{SU}(2)$. Recalling Eqns. (1.6), (1.7), both $\hat{H}_V$ and $\hat{N}$ are $\text{SU}(2)$ symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) $\text{SU}(2)$ symmetric.

Note that the $U^c(1)$ gauge symmetry and the $\text{SU}(2)$ spin rotations symmetry can be combined,

$$U^c(1) \otimes \text{SU}(2) = \text{U}(2)$$

We will refer to $\text{U}(2)$ symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z ,

$$U^z(1) \subset \text{SU}(2)$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the “real” z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the $\text{SU}(2)$ generators, starting from $\hat{S}^z$, after we have chosen the direction z . In terms of commutation relations, a given operator can respect $U^z(1)$ but not $\text{SU}(2)$: in that case, it will commute to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 1.2, representing for a given vector $\mathbf{m}$ the passage from full $\text{SU}(2)$ rotational symmetry around any direction to reduced $\text{U}(1)$ rotational symmetry around just one direction.

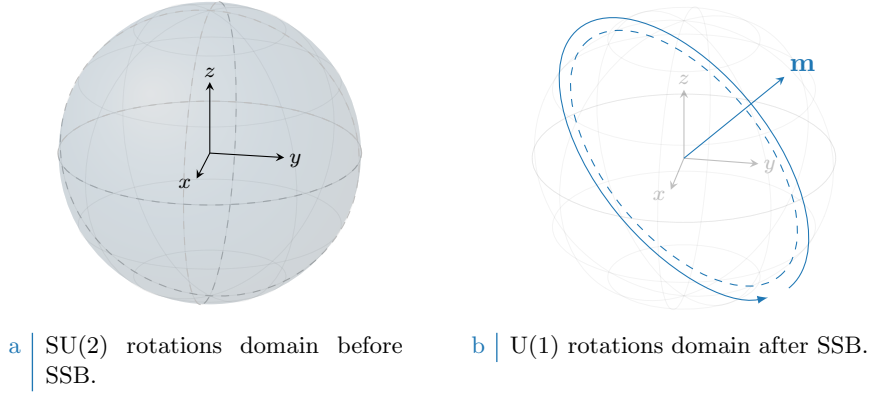


Figure 1.2 | Graphic representation of the concept of $SU(2) \rightarrow U^z(1)$ SSB. “Before” SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). “After” SSB we have symmetry only for rotations around some vector $\mathbf{m}$.

1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary operation

$$P : \hat{c}_{i\sigma} \rightarrow e^{i\eta_{i\sigma}} \hat{h}_{i\sigma}^\dagger \quad (1.9)$$

being in general η_i a phase dependent on the fermionic quantum state ($i\sigma$). Inserting the above transformation in the EHM, we get

$$P\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^\dagger + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\uparrow}^\dagger - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger$$

For the two interactions $\hat{H}_U, \hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (??),

$$P\{\hat{H}\} \stackrel{?}{=} -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument of Arovas et al. [1] about the conventional HM, let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned} \{\hat{h}_{i\sigma}, \hat{h}_{j\sigma'}^\dagger\} &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \{\hat{c}_{i\sigma}^\dagger, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \delta_{ij} \delta_{\sigma\sigma'} \\ &= \delta_{ij} \delta_{\sigma\sigma'} \end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma} \hat{h}_{i\sigma}^\dagger = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ . It follows, for quartic interactions

$$\begin{aligned} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger &= -\hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \\ &= \hat{h}_{i\sigma} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} - \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger \delta_{ij} \delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma} \hat{h}_{j\sigma'} - \hat{h}_{i\sigma}^\dagger \hat{h}_{i\sigma} + \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger - \left(\hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \right) \delta_{ij} \delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma} \hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \right) \delta_{ij} \delta_{\sigma\sigma'} \quad (1.10) \end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^{\dagger}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma} + \pi)} \hat{h}_{i\sigma}^{\dagger} \hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k - 1)\pi \quad \text{with} \quad k \in \mathbb{Z} \quad (1.11)$$

valid for NN sites.

Local repulsion. Substitute Eq. (1.10) in $\hat{H}_U$,

$$\begin{aligned} P\{\hat{H}_U\} &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^{\dagger} \hat{h}_{i\uparrow}^{\dagger} \\ &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^{\dagger} \hat{h}_{i\downarrow}^{\dagger} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_{i \in \mathcal{L}} (1) \\ &= U \sum_i \hat{h}_{i\uparrow}^{\dagger} \hat{h}_{i\downarrow}^{\dagger} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U(\hat{N} - L_x L_y) \end{aligned}$$

For the half-filled state $\langle \hat{N} \rangle = L_x L_y$. The unitary map of Eq. (1.9), independently of the condition of Eq. (1.11), leaves the hamiltonian identical at half-filling.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned} P\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^{\dagger} \hat{h}_{i\sigma}^{\dagger} \\ &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^{\dagger} \hat{h}_{j\sigma'}^{\dagger} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\ &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^{\dagger} \hat{h}_{j\sigma'}^{\dagger} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y) \end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned} \sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\ &= \frac{L_x L_y}{2} \times z \end{aligned} \quad (1.12)$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (1.9). The condition of Eq. (1.11) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$P : \begin{cases} \hat{c}_{i\sigma} \rightarrow \hat{h}_{i\sigma}^{\dagger} & \text{for } i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \rightarrow -\hat{h}_{i\sigma}^{\dagger} & \text{for } i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by π on the two sublattices, is good enough. As for the conventional HM [1], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 1.1 all identified symmetries.

Symmetry	Operations
$T_1^{(x)} \otimes T_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$U^c(1)$	Charge-conserving global phase shifts
$SU(2)$	Three-dimensional rotations in global spin space

Table 1.1 | Model symmetries and relative group operations. Note that $U^c(1)$ represents the gauge symmetry linked to charge conservation.

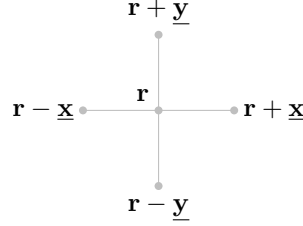


Figure 1.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

1.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some SSB. In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. ???. The ending point were Eqns. (??), (??), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\text{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $SU(2) \rightarrow U^z(1)$ and $T_1^{(x)} \otimes T_1^{(y)} \rightarrow T_2^{(x)} \otimes T_2^{(y)}$ SSBs. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $U^c(1) \rightarrow 0$ and $C_4 \rightarrow C_2$ SSBs. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

Nematic effects are observed in unconventional superconductors, and this is the reason of our interest in rotational SSB. We will theoretically discuss the nematic SSB $C_4 \rightarrow C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \rightarrow \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \quad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.3.

1.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

In order to obtain the self-consistent values of $h_{\alpha\beta}^{+-}$, $h_{\alpha\beta}^{++}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (??),

$$\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}} \quad (1.13)$$

where the equal sign holds as long as we perform the averaging operation on the subspace of **gaussian states**. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle$$

Recalling the groups of Tab. 1.1, we see that the density operator preserves all symmetries. Single-site density operators commute with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} = \hat{S}_{\mathbf{r}}^+$$

being $S_{\mathbf{r}}^+$ the local raising operator. From SU(2) algebra we have:

$$[\hat{S}_{\mathbf{r}}^+, \hat{S}^\alpha] \neq 0 \quad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (1.5).

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (1.4) to $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rightarrow e^{2i\eta} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. To prove this, it suffices to compute commutators of $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow}] \hat{c}_{\mathbf{r}\downarrow}^\dagger - \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\downarrow}] \end{aligned}$$

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.2 Summary of selection rules for the Wick's decomposition of Eq. (1.13) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

Since:

$$\begin{aligned}
[\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma}^\dagger \\
&= -\hat{c}_{\mathbf{r}\sigma}^\dagger (1 - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}) \\
&= -\hat{c}_{\mathbf{r}\sigma}^\dagger
\end{aligned} \tag{1.14}$$

we have:

$$[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] = -\frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger + \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger = 0$$

Moreover, considering the raising operator:

$$\begin{aligned}
[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^+] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} \hat{c}_{\mathbf{r}'}^\dagger \hat{c}_{\mathbf{r}'} \right] \\
&= [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}] \\
&= [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger] \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} - (\hat{c}_{\mathbf{r}\uparrow}^\dagger)^2 [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\downarrow}] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutes with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 1.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (1.13), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

1.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\hat{H}_V \equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}}$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} f(\mathbf{r}, \mathbf{r}') = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} f(\mathbf{r}, \mathbf{r} + \boldsymbol{\delta}) \tag{1.15}$$

with f some function. We are using the notation of Fig. 1.3. The symbol “ $\mathcal{L}/2$ ” indicates one of the two sublattices $\mathcal{S}_1, \mathcal{S}_2$ represented in Fig. [Missing]. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'})$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\begin{aligned}\hat{H}_V^{\uparrow\downarrow} &= -V \sum_{\mathbf{r} \in \mathcal{S}_1} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\delta\downarrow}) \\ \hat{H}_V^{\downarrow\uparrow} &= -V \sum_{\mathbf{r} \in \mathcal{S}_1} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}+\delta\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}-\delta\uparrow})\end{aligned}$$

These two operators are “complementary”: $\sigma = \uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\delta\downarrow})\end{aligned}\quad (1.16)$$

Wick’s theorem applied to these quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\delta\downarrow} \rangle &= \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}+\delta\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \\ &\simeq \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}+\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}+\delta\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\delta\downarrow} \rangle \langle \hat{c}_{\mathbf{r}+\delta\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\delta\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{r}+\delta\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}}\end{aligned}\quad (1.17)$$

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\sigma \in \{\uparrow, \downarrow\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma})\end{aligned}\quad (1.18)$$

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick’s theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma} \rangle &= \langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \\ &\simeq \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}_{\mathbf{r}+\delta\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma} \rangle \langle \hat{c}_{\mathbf{r}+\delta\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma}^\dagger \rangle \langle \hat{c}_{\mathbf{r}+\delta\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Cooper}}\end{aligned}\quad (1.19)$$

Hartree terms. As it was for $\hat{H}_U$, Hartree EVs of Eqns. (1.16), (1.18) are not subject to selection rules for any symmetry of Tab. 1.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{S}^z] &= \left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] \hat{c}_{\mathbf{r}+\delta\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \hat{c}_{\mathbf{r}\sigma}^\dagger [\hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{n}_{\mathbf{r}+\delta\sigma}] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \right) \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\delta\sigma'}\end{aligned}\quad (1.20)$$

where in last passage we used Eq. (1.14) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\delta\sigma'}, \hat{n}_{\mathbf{r}+\delta\sigma}] = \hat{c}_{\mathbf{r}+\delta\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (1.20) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $U^z(1)$.

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.3 | Summary of selection rules for the Wick's decomposition of Eqns. (1.17), (1.19) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Cooper terms. For the Cooper terms, identical considerations as for $\hat{H}_U$ hold. Gauge symmetry must be broken to allow for non zero EVs, while SU(2) symmetry survives. The proof is analogous to the one given for $\hat{H}_U$.

In Tab. 1.3 we summarise these considerations, giving the selection rules on the EVs of Eqns. (1.17), (1.19), dividing in the two sectors “o.s.” and “s.s.”. By identifying these selection rules, we set the ground for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

1.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{-i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \quad (1.21)$$

Throughout the text, $\stackrel{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. “BZ” is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i\mathbf{k} \cdot \mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \quad (1.22)$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}=\mathbf{0}} \quad (1.23)$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. Let us transform $\hat{H}_t$:

$$\begin{aligned} \hat{H}_t &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}] \\ &\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\ &= -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \times \sum_{\delta \in \{\mathbf{x}, \mathbf{y}\}} e^{i\mathbf{k}' \cdot \delta} \\ &= -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2} \right] \end{aligned} \quad (1.24)$$

$$= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (1.25)$$

where we used Eqns. (1.15), (1.23). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t (\cos k_x + \cos k_y) \quad (1.26)$$

Local repulsion. The local repulsion transforms as:

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \\ &\stackrel{\mathcal{F}}{=} U \sum_{\mathbf{r} \in \mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i\mathbf{k}_1 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_1\uparrow}^\dagger e^{i\mathbf{k}_2 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_2\downarrow}^\dagger e^{-i\mathbf{k}_3 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_4\uparrow} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^\dagger \hat{c}_{\mathbf{k}_2\downarrow}^\dagger \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^\dagger \hat{c}_{\mathbf{k}_2\downarrow}^\dagger \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \delta_{\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4} \end{aligned}$$

where we used Eq. (1.22). Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}, \mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{k}_2 \equiv \mathbf{K} - \mathbf{k} \quad \mathbf{k}_3 \equiv \mathbf{K} - \mathbf{k}' \quad \mathbf{k}_4 \equiv \mathbf{K} + \mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (1.27)$$

Non-local attraction. To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma'})$$

with:

$$\hat{H}_V = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r} \pm \delta\sigma'} &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma'}^\dagger \hat{c}_{\mathbf{r} \pm \delta\sigma'} \hat{c}_{\mathbf{r}\sigma} \\ &\stackrel{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} e^{\pm i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

Then:

$$\begin{aligned} h_{\mathbf{r}\sigma\sigma'} &\stackrel{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \\ &\quad \times \left(e^{i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} + e^{-i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \right) \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \cos[(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta] \hat{c}_{\mathbf{k}_1\sigma}^\dagger \hat{c}_{\mathbf{k}_2\sigma'}^\dagger \hat{c}_{\mathbf{k}_3\sigma'} \hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (1.23) gives back momentum conservation

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4}$$

where we used Eq. (1.23). Define $\mathbf{k}, \mathbf{k}'$ and $\mathbf{K}$ as we did for the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k} - \mathbf{k}'$$

Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\stackrel{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \cos(\delta\mathbf{k} \cdot \boldsymbol{\delta}) \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma'}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma'}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \quad (1.28)
\end{aligned}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}
\hat{H}_V &\stackrel{\mathcal{F}}{=} \overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\quad - \underbrace{\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}_{\mathbf{K}-\mathbf{k}\bar{\sigma}}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\bar{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \rightarrow -\mathbf{k} \quad \mathbf{k}' \rightarrow -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (1.29)$$

The $\mathcal{F}$ -transform of the s.s. part is given explicitly by:

$$\begin{aligned}
\hat{H}_V^{(\text{s.s.})} &= -\frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \\
&\quad \times \left[\hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}_{\mathbf{K}+\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow} \right] \quad (1.30)
\end{aligned}$$

All these expressions are going to be used in the analysis of each phase. In reciprocal space, for a quartic operator $\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$ we distinguish contractions as follows:

- Hartree contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Hartree}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overbrace{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Hartree}}$$

- Fock contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Fock}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overbrace{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Fock}}$$

- Cooper contractions:

$$\overbrace{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Cooper}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \overbrace{\hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}^{\text{Cooper}}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.

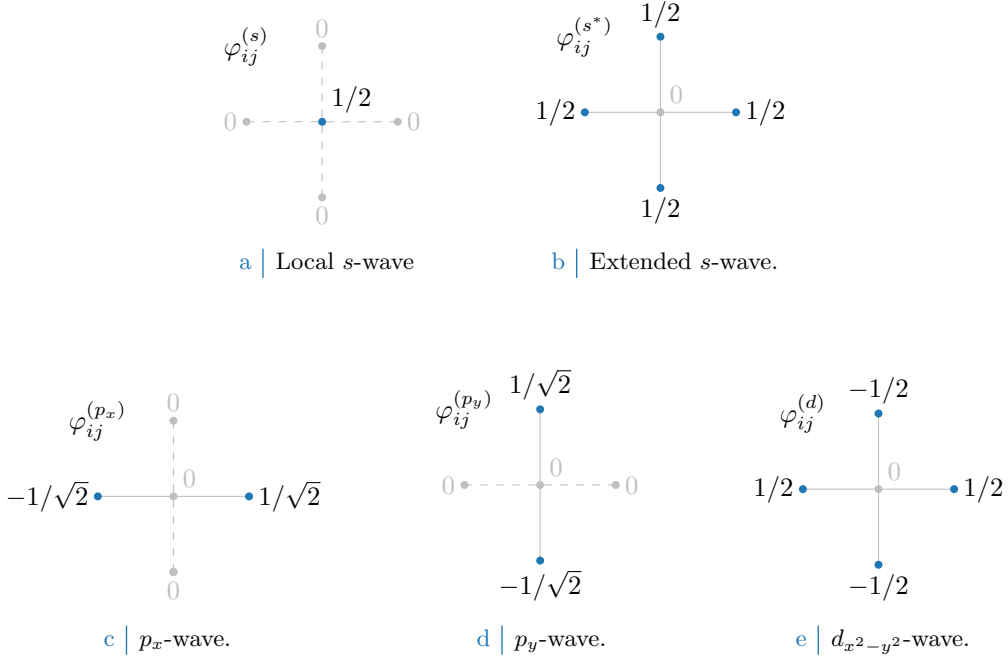


Figure 1.4 Form factors at different topologies, as listed in Tab. 1.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for φ_{δ} , while dashed lines represent vanishing factors.

1.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \iff |\mathbf{r} - \mathbf{r}'| = 0, 1$$

in units of the lattice step. The following identity holds:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \delta_{\mathbf{r}' = \mathbf{r} + \underline{\mathbf{x}}} \\ &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \end{aligned}$$

where we defined the set of orthonormal functions $\varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$ listed explicitly in Tab. 1.4. The orthonormality condition reads:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= f(\mathbf{r}, \mathbf{r}') \varphi_{\mathbf{r}\mathbf{r}'}^{(s)} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \end{aligned}$$

Structure	Form factor	Graph
s -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s)} = \delta_{\mathbf{r}\mathbf{r}'}$	Fig. 1.4a
Extended s -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 1.4b
p_x -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}]$	Fig. 1.4c
p_y -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 1.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(d)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 1.4e

Table 1.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 1.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

Structure	Structure factor	Graph
s -wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 1.4a
Extended s -wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 1.4b
p_x -wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2} \sin k_x$	Fig. 1.4c
p_y -wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2} \sin k_y$	Fig. 1.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 1.4e

Table 1.5 | Structure factors derived from the correlation structures of Tab. 1.4. The functions hereby defined are orthonormal, and are plotted in Fig- 1.5.

Thus, defining:

$$\begin{aligned}
f^{(s)} &\equiv f(\mathbf{r}, \mathbf{r}) \\
f^{(s^*)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\
f^{(p_x)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) \\
f^{(p_y)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\
f^{(d)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}})
\end{aligned}$$

we get the decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$$

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$ -transformed functions of Tab. 1.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

We will refer to these harmonics as “symmetry channels”, indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations. To distinguish in symmetry channels is useful to apply selection rules to the topology of the order parameter based on the planar rotational symmetry of the studied phase.

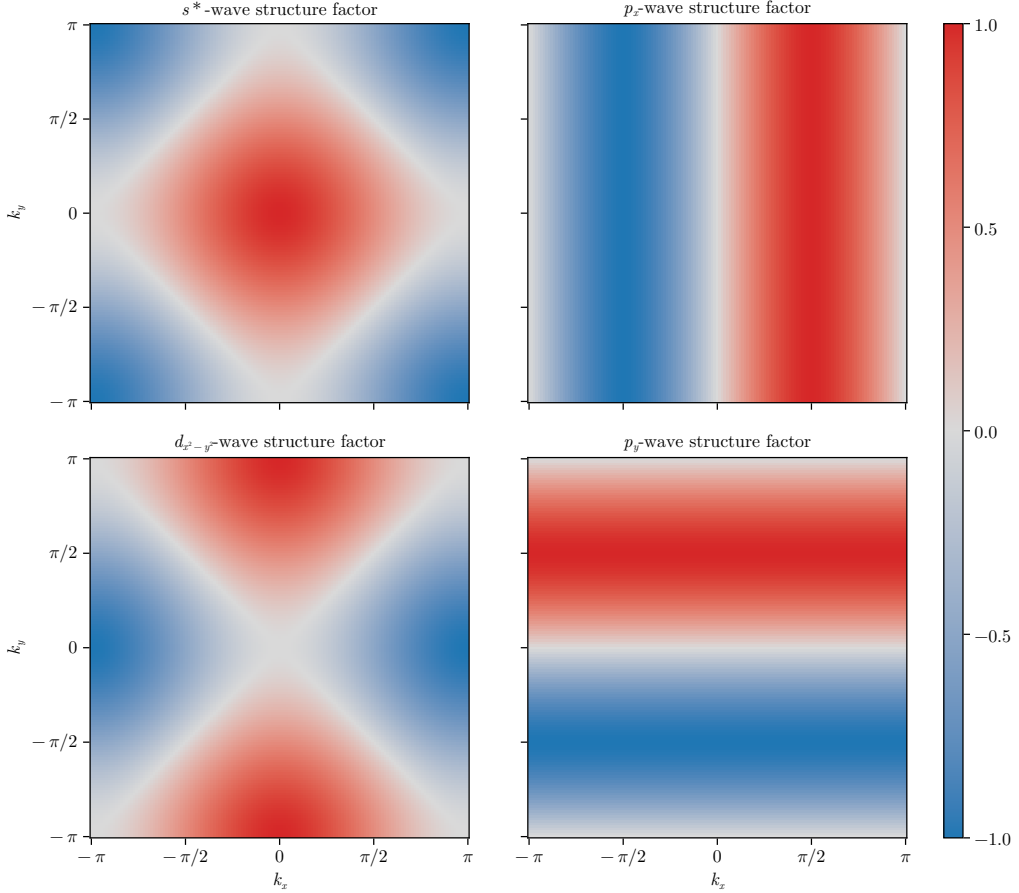


Figure 1.5 Representation of the various structure factors listed in Tab. 1.5 on the BZ. For the s^* and $d_{x^2-y^2}$ -wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (1.28), where is present the factor $\cos(\delta k_x) + \cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$). Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \quad s_\ell \equiv \sin k_\ell \quad c'_\ell \equiv \cos k'_\ell \quad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{(c_x c'_x + c_y c'_y)}_{\text{Symmetric}} + \underbrace{(s_x s'_x + s_y s'_y)}_{\text{Anti-symmetric}} \quad (1.31)$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \rightarrow (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}, \mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) &= \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
&+ s_x s'_x && (p_x\text{-wave}) \\
&+ s_y s'_y && (p_y\text{-wave}) \\
&+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \quad (1.32)$$

$$= \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \quad (1.33)$$

This decomposition will become particularly handy in later calculations.

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